

XIAOTIAN LIU

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EDUCATION

H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology *Atlanta, USA*
Ph.D. Candidate, Advised by Prof. Christos Alexopoulos and Prof. Edwin Romeijn *Sept. 2023 - 2027 (expected)*

School of Electronics Engineering and Computer Science, Peking University *Beijing, China*
B.S. in Electronics Engineering *Sept. 2018 - June 2022*

Guanghua School of Management, Peking University *Beijing, China*
B.S. in Management *Sept. 2019 - June 2022*

RESEARCH INTERESTS

My research focuses on pushing the frontier of operations management (OM) through deep neural networks, the key driver of the current AI wave. I investigate how to effectively train deep neural networks (DNNs) for modern OM problems involving high-dimensional data and nonlinear dynamics. I also work on combining DNNs with operational knowledge and exploiting advanced DNN architectures, such as transformers, for intelligent OM decision-making.

PUBLICATIONS

(^athesis advisor, ^bstudent collaborator, ^cprofessor collaborator, ^dthesis co-advisor)

[1] Xiaotian Liu, Christos Alexopoulos^a, Yijie Peng^c, and Cheng Zhang^c, “Production Planning with Markovian Production Relationships,” *Production and Operations Management* (UTD24), 2026, <https://journals.sagepub.com/doi/10.1177/10591478261445695>

[2] Xiaotian Liu, Ming Hu^c, Yijie Peng^c, and Yaodong Yang^c, “Multi-Agent Deep Reinforcement Learning for Multi-Echelon Inventory Management,” *Production and Operations Management* (UTD24), 2025, <https://journals.sagepub.com/doi/full/10.1177/10591478241305863>

- *Production and Operations Management* 2023–2026 Top-20 Most Cited Paper (Journal Official Ranking)
- The Only AI-Related Paper in The Top-20 Citation List

[3] Xiaotian Liu, Yijie Peng^c, Gongbo Zhang^b, and Ruihan Zhou^b, “An Efficient Node Selection Policy for Monte Carlo Tree Search with Neural Networks,” *INFORMS Journal on Computing* (UTD24), 2025, <https://pubsonline.informs.org/doi/full/10.1287/ijoc.2023.0307>

- *INFORMS Journal on Computing* Meritorious Paper Award (The Only Awardee in 2024)
- The First Time This Award Was Presented to a Chinese Research Team

[4] Xiaotian Liu, Christos Alexopoulos^a, and Yijie Peng^c, “A Simulation-Driven Machine Learning Framework for Large-Scale Inventory Management,” *Annals of Operations Research* (ABS3), 2025, <https://link.springer.com/article/10.1007/s10479-025-06857-5>

[5] Qinghao Wang^b, Jinyang Jiang^b, Xiaotian Liu, Tao Ren^b, Yi Zheng^b, Cheng Zhang^c, Yaodong Yang^c, and Yijie Peng^c, “A Unified Framework for Multi-Stage Decision Optimization with Deep Reinforcement Learning and Foundation Models,” *2025 IEEE 21st International Conference on Automation Science and Engineering (CASE)*, 2025, <https://ieeexplore.ieee.org/abstract/document/11163883>

[6] Gongbo Zhang^b, Xiaotian Liu, and Yijie Peng^c, “A Simulation Optimization Method for Scheduling Automated Guided Vehicles in a Stochastic Warehouse Management System,” *2023 Winter Simulation Conference (WSC)*, 2023, <https://ieeexplore.ieee.org/abstract/document/10407793>

- Second Place Winning Method in *2022 Winter Simulation Conference* Case Study Competition

[7] Xiaotian Liu, Lei Yang^b, Xiaoyu Zhang^b, and Xiaohui Duan^c, “MA-ResNet50: A General Encoder Network for Video Segmentation,” *2022 International Conference on Computer Vision Theory and Applications (VISAPP)*, 2022, <https://pdfs.semanticscholar.org/3d6f/8f0a026029521815c6402f6c52f937fc7196.pdf>

WORKING PAPERS

[8] (Job Market Paper) Ming Hu^c, Xiaotian Liu, and Guohao Shen^c (authors listed in alphabetical order), “AI as Flesh, OM as Bone: Semiparametric Regression with Deep Neural Networks,” Under revision (previous decision: reject and resubmit) at *Manufacturing & Service Operations Management* (UTD24), 2026, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5460214

[9] Xiaotian Liu, Christos Alexopoulos^a, and Edwin Romeijn^d, “Learning Meets Convex Structure: Offline Approximation for Inventory Control,” Under review at *Production and Operations Management* (UTD24), 2026, https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5393319

- Appeared in 2025 SSRN Top-10 Download List for Industrial & Manufacturing Engineering

[10] Xiaotian Liu, Christos Alexopoulos^a, and Edwin Romeijn^d, “Adaptive Deployment of Black-Box Inventory Policies in Nonstationary Environments,” Work in progress, 2026

PATENTS

[1] Yijie Peng, Xiaotian Liu, Gongbo Zhang, Cheng Zhang, and Min Zhan, “An Adaptive Bitrate Control Method and System,” Chinese Invention Patent No. ZL202510949984.6, 2025.

HONORS & AWARDS

INFORMS Journal on Computing Meritorious Paper	2025
Second Place Award (ranking 2/31), Winter Simulation Conference 2022 Case Study Competition	2022
Mao Yugang Scholarship for Outstanding Undergraduate in Research (Top 1%, Peking University)	2022
Outstanding Undergraduate Research Program (Top 1%, Peking University)	2021

TEACHING

Deterministic Optimization (ISYE6669)	Georgia Institute of Technology, USA
Teaching Assistant	Spring 2025
Computational Data Analysis (ISYE6740)	Georgia Institute of Technology, USA
Teaching Assistant	Summer 2024, Fall 2024, Summer 2025

PRESENTATIONS / INVITED TALKS

A Unified Framework for Decision Optimization with Deep Reinforcement Learning and Foundation Models	
2025 International Conference on Automation Science and Engineering	Los Angeles, California, USA, 2025
A Simulation-Driven Machine Learning Framework for Large-Scale Inventory Management	
2023 INFORMS Annual Meeting	Phoenix, Arizona, USA, 2023
Second Place Winner Presentation for WSC Data Competition	
2022 Winter Simulation Conference	Singapore, 2022
MA-ResNet50: A General Encoder Network for Video Segmentation	
2022 International Conference on Computer Vision Theory and Applications	Online, 2022

ACADEMIC SERVICE

Paper Reviewer: INFORMS Journal on Computing; Winter Simulation Conference